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AQCat25: unlocking spin-aware, high-fidelity machine learning potentials for heterogeneous catalysis

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Large-scale datasets have enabled highly accurate machine learning interatomic potentials (MLIPs) for general-purpose heterogeneous catalysis modeling. There are, however, some limitations in what can be treated with these potentials because of gaps in the underlying training data. To extend these capabilities, we introduce AQCat25, a dataset of 13.5 million density functional theory (DFT) single-point calculations designed to enhance the treatment of systems where spin polarization and/or higher fidelity are critical. We also investigate integrating datasets, such as AQCat25, with the broader Open Catalyst 2020 (OC20) dataset to create spin-aware models without sacrificing generalizability. We find that directly tuning a general model on AQCat25 leads to catastrophic forgetting of the original dataset's knowledge. Conversely, joint training strategies prove effective for improving accuracy on new distributions without sacrificing general performance. This joint approach introduces a challenge, as the model must learn from a dataset containing both mixed-fidelity calculations and mixed-physics (spin-polarized vs. unpolarized). We show that explicitly conditioning the model on this system-specific metadata, for example, by using Feature-wise Linear Modulation (FiLM), successfully addresses this challenge and further enhances accuracy. Ultimately, we establish an effective protocol for bridging DFT fidelity domains to advance the predictive power of foundational models in catalysis.

Over the past three decades, computational approaches that couple first-principles density functional theory (DFT) with microkinetic modeling have become a cornerstone of modern heterogeneous catalysis research by providing a framework for rational catalyst design[1–5]. Numerous studies have linked atomic-scale surface chemistry to macroscopic kinetic observables, enabling the elucidation of complex reaction mechanisms for a variety of critical industrial heterogeneous catalytic processes, including, but not limited to, ammonia synthesis[6,7], methanol synthesis[8,9], Fischer-Tropsch synthesis[10,11], selective hydrogenation[12,13], steam reforming of methane[14,15], the water-gas shift reaction[16,17], and ethylene epoxidation[18,19]. Many of these studies have leveraged unifying concepts such as d-band theory and the Brønsted-Evans-Polanyi relations[20–23], which correlate the binding and transition-state energies of elementary reactions, facilitating the construction of volcano plots that predict optimal catalyst performance, in some cases even leading to experimentally validated discovery of new catalysts[24,25]. Despite these successes, the prohibitive cost of DFT largely limits its application to relatively simple networks of reactions taking place over idealized low-index facets of unary and binary catalyst materials[2,23].

Machine learning interatomic potentials (MLIPs) have emerged as an attractive alternative to estimate electronic structure properties at near-quantum accuracy for a small fraction of the computational cost[26–31]. These models learn the interactions required to predict the potential energy landscape of atomistic systems from large-scale public databases of DFT calculations[32–35]. State-of-the-art models for heterogeneous catalysis are made possible by Meta FAIR's Open Catalyst 20, 22, and 25 datasets[36–38], which collectively consist of nearly 300 million single-point DFT calculations of adsorbate-surface interactions relevant for reactions of carbon, hydrogen, oxygen, and nitrogen over a diverse catalyst space spanning most

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of the periodic table. The introduction of machine learning methods into computational catalysis workflows has begun to enable studies of reaction network complexity[39–42] and catalyst structural dynamics[43–45] that were completely inaccessible just 10 years ago.

Although the sheer scale of these datasets has necessitated some compromises in the fidelity of the underlying training data, convergence of the resulting adsorption energies benefits from error cancellation and has been validated with respect to most DFT settings, with plane-wave cutoff and smearing width requiring some improvements to accurately capture total energies of non-metals[46]. One of the most significant gaps in existing large-scale heterogeneous catalysis datasets is the treatment of magnetism. Since spin-polarized DFT calculations are considerably more expensive than spin-unpolarized calculations, spin is often omitted in the interest of scale and throughput[47]. The consequence of this choice is that the resulting models are not suitable for many industrially relevant catalytic processes, such as ammonia synthesis[6] and Fischer-Tropsch synthesis[10], which rely on earth-abundant first-row transition metals (e.g., iron, cobalt, and nickel) that exhibit especially strong spin polarization effects on binding energies and activation barriers[48–51]. As the field moves towards discovering new, low-cost, and sustainable catalytic materials to replace precious metals, the importance of treating magnetic effects when training foundational MLIPs becomes increasingly paramount.

Alongside progress in data generation, developments in machine learning architectures, often based on equivariant graph neural networks, have improved performance on atomistic tasks. For heterogeneous catalysis, models like eSEN[52], EquiformerV2[53] and EScAIP[54] have achieved state-of-the-art results. A significant leap towards broader universality is the Universal Model for Atoms (UMA)[55], trained on ~ 500 million structures across diverse chemical domains (molecules, materials, catalysts). UMA modifies the eSEN architecture with additive embeddings for global context (charge, spin, DFT task) and a Mixture of Linear Experts (MoLE) routed by this context plus element composition. UMA's core design goal is to accurately reproduce the original physics of each training task (e.g., spin-unpolarized OC20), operating as a multi-task surrogate rather than a model explicitly designed to perform cross-fidelity corrections between different levels of theory. A distinct advantage of total energy models like UMA is their ability to better capture, among other effects, the restructuring of bare catalyst slabs[46].

Other approaches have focused on integrating low- and high-fidelity data for related tasks, such as by augmenting node features with a fidelity one-hot encoding and applying both common and fidelity-specific weights in modified linear layers[56] or utilizing a model's intrinsic global state feature to embed fidelity context during message passing[35]. For example, Ko and Ong demonstrated that a single multi-fidelity model trained with a small fraction of high-fidelity data could achieve similar accuracy to a single-fidelity model requiring eight times the amount of costly high-fidelity training data[35]. Alternatively, other methods utilize architectural separation, such as dynamically using separate prediction heads branching from a shared backbone for each fidelity level[57,58].

While recent work has produced model architectures that can incorporate spin[47,59–64], their predictive power is limited by the absence of high-quality, spin-polarized training data for heterogeneous catalysis. Given the success of methods for adapting models to new data[37,65–69], alongside advancements in training universal models from diverse datasets, we see a clear opportunity to develop improved foundational models specifically for spin-polarized, high-fidelity catalytic systems.

Here, we present the AQCat25 dataset and baseline AQCat25-EV2 models (Fig. 1), which improve upon the performance of EquiformerV2-31M and EquiformerV2-153M adsorption energy MLIPs for heterogeneous catalysis in three key ways: increasing the fidelity of the reference DFT calculations, explicitly incorporating spin polarization for magnetic elements, and introducing new elements to the model domain that are underrepresented in existing datasets. Through this work, we demonstrate data-efficient methodologies for building multi-fidelity MLIPs that span distinct physical regimes (such as spin polarization) to new domains of chemistry while ensuring that the models maintain accuracy and generalizability across a wide range of catalysts and reactions. The dataset, models, and code are available publicly to support further developments by the academic community.

Results

Dataset composition

Summary statistics of the AQCat25 dataset and comparisons to the Open Catalyst 2020 dataset (OC20 dataset) are shown in Fig. 2 and Supplementary Fig. S1. As shown in Fig. 2b, there is a significantly higher proportion of non-metal systems and a lower proportion of metal systems in the AQCat25 dataset compared to the OC20 dataset. The proportions of the other two categories (non-metal and metalloid, and metalloid) are roughly equal. Because non-metal systems typically have poorer performance compared to intermetallics[70], we will examine key model performance metrics split over these material categories. The adsorption energy and maximum force distributions (Fig. 2c, d) reveal that the AQCat25 dataset is biased towards higher force, higher energy systems when compared to OC20. This can be explained by the more aggressive approach taken when sampling high-force systems. Here, we used larger standard deviations to sample rattled configurations and also included high-energy transition state-like systems.

Optimizing data generation strategies for fine-tuning

We wanted to explore opportunities to improve our data generation strategy to maximize model performance while minimizing the cost associated with dataset generation and model fine-tuning. To do this, we looked at the change in model performance as a function of two variables: the number of DFT single points seen per system and the number of slabs. For heterogeneous catalyst systems, there are two types of diversity the models must generalize across: (1) the adsorbates and (2) the material surfaces the adsorbates are adsorbed to. The latter is much more complex. We initially adopted a scheme of performing four adsorbate-slab relaxations per slab to reduce the number of slab relaxations that needed to be performed, as this data is not directly used in model training for referenced energy models. This turned out to be a suboptimal choice, however, because material diversity is important to tackle. For the future, we would choose to perform one adsorbate-slab calculation per slab. In Fig. 3, we show the relationship between model force performance, number of slabs seen, and number of DFT single points seen per system. Models were directly tuned starting from the publicly available 31M parameter EV2 model (All + MD)[53].

The amount of data used to train the models directly impacts the cost to train and to iterate between architectures and ablations. Gasteiger and colleagues have shown the OC20-2M subset to be representative of the full OC20 dataset, primarily because it preserved the underlying chemical diversity[71]. Other approaches use more complex stratified sampling (based on feature-space clustering) to ensure that diverse, high-energy, and uncommon configurations are explicitly captured to improve model robustness[72]. Given the precalculated training data, we explored the opportunity to reduce model training costs by sampling the frames along adsorbate-slab relaxation trajectories. Sampling was performed using force-stratified selections of the trajectory to obtain a representative distribution of systems. For consistency, models were trained for a nearly constant number of total gradient updates, approximately equivalent to the number of steps in one epoch of the largest split. Further, the data ablation for sampling frames from trajectories (Fig. 3a) was designed to probe redundancy in highly autocorrelated data and thus only applied to the relaxations and molecular dynamics (MD) data; the rattled and transition state data were included entirely for each model. To enable a cleaner evaluation of the data cost-benefit trade-off for tuning, without confounding the results with the model's ability to learn new, unseen elements, we restricted the subsampling experiments to AQCat25 systems with elements already seen by the 31M pretrained model. Figure 3a reports the validation mean absolute error (MAE) from the final training checkpoint, which highlights the risk of overfitting. Conversely, Fig. 3b plots the MAE from the best-performing model during training for each slab count (averaged over all k values) against

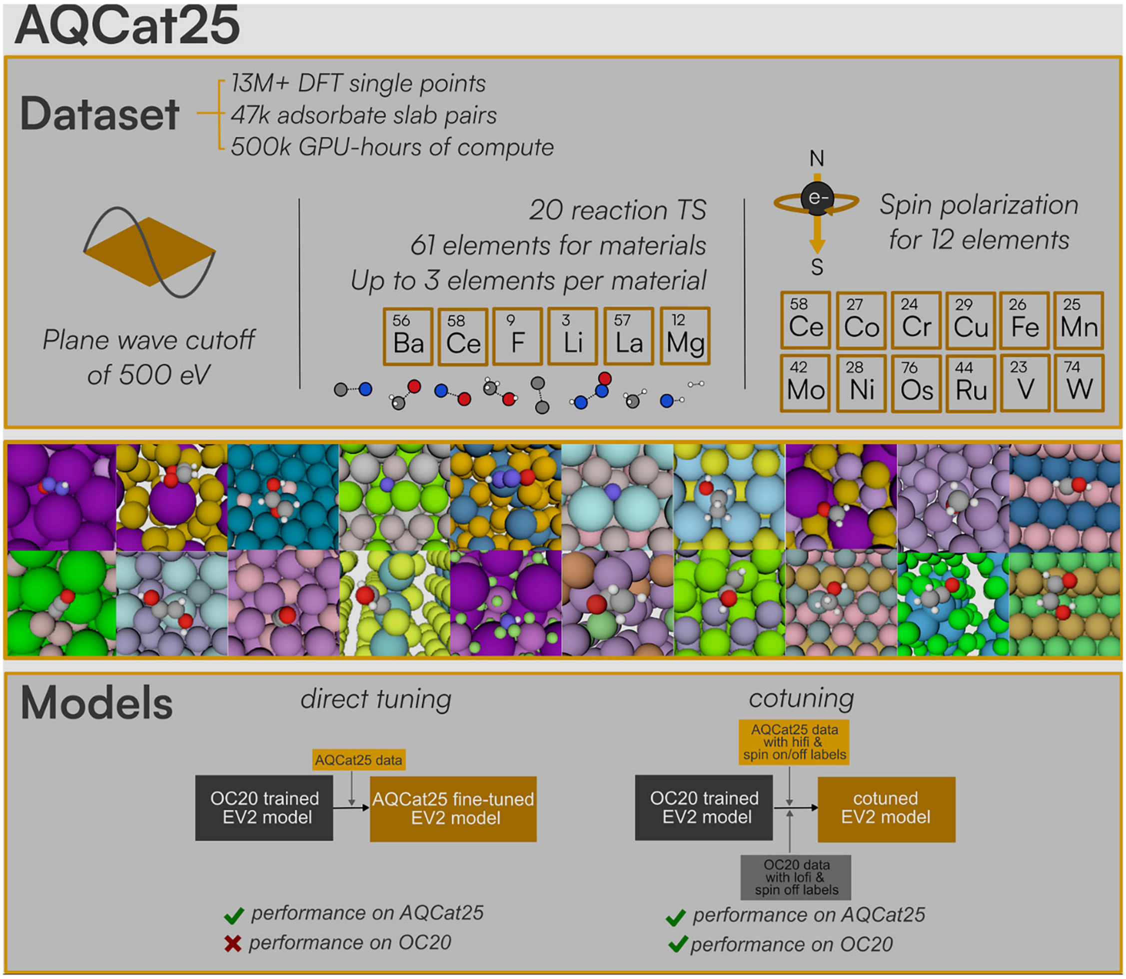

Fig. 1 | A summary of the AQCat25 dataset and models. Spin polarization has been included for 12 important elements. A plane wave cutoff of 500 eV is used. Six new elements are added when compared to the OC20 dataset, as well as 20 transition-state adsorbates. Models were jointly tuned/trained from scratch on OC20 and AQCat25 to achieve good performance on their validation and test splits, respectively.

the data generation cost. As shown in Fig. 3a, for small numbers of slabs, sampling a subset of frames rather than using the full trajectory actually leads to better force MAE values. This is likely due to a high propensity for overfitting, which is shown in Supplementary Fig. S2 and remedied by sampling. At larger slab numbers, the differences are minor, but the cost to train will be lower if sampling is performed. For energy MAE, there is no substantial trend with changing sample size (see Supplementary Fig. S3). Therefore, sampling frames are a useful cost-saving strategy. We adopted a subsampling approach for the model adaptation experiments presented in subsequent sections. However, for those experiments, we employed random sampling per trajectory rather than the force-stratified approach. We found this yielded slightly improved performance, likely due to increasing the representation of low-force, near-equilibrium frames that are critical for downstream adsorption energy tasks.

Unsurprisingly, having more unique slabs in the dataset improves performance. However, there is a cost trade-off to be made, which is explored in Fig. 3b. Comparing 250–1000 and 1000–4000, there is a large improvement in the metrics. Going from 4000 to 10,000 slabs, however, we are beginning to enter the domain of having diminishing returns on our computational investment, indicating that the number of slabs we calculated in this dataset was a reasonable choice. Nonetheless, this experiment primarily assessed convergence with respect to the number of unique slabs, not the total number of unique adsorbate-catalyst combinations, which warrants further investigation. Here, the additional compute cost is referenced to the 250-slab dataset. This value serves as a proxy for the total computational investment, which is expected to correlate with the true data generation cost and illustrates the trend of diminishing returns.

The apparent redundancy revealed by randomly sampling offers a potential opportunity: what if, instead of optimizing full relaxation trajectories, we only calculate the first k points? This could greatly reduce the computational cost to generate the data, but it introduces a potential new problem. It biases the relaxation data towards higher force states, which could cause models trained on the data to have poor performance on low force systems. To investigate this, we divided the validation set into force percentiles and examined changes in performance on the different percentiles. As a baseline, we first assessed the pretrained model's performance in Fig. 3c. Performance decreases with increasing force percentile, with the exception of the highest force percentile considered, which has better performance than even the lowest force percentile. This is likely a reflection of the underlying OC20 dataset that contains MD, rattled systems, and relaxations. MD and rattled data have high forces, while relaxations contain many frames in the low force regime. Figure 3d shows the evolution of model performance on the force percentile segmented validation split with an increasing number of frames sampled. Please note that here the data ablation also only applied to relaxations; all models were trained on the transition state (TS)-like and rattled data, but none included the MD data. Performance overall improves with the number of frames sampled, but it does not occur in a way that disproportionally affects specific force segments from 0 to 99%. The one exception to this is the highest force percentile, which modestly improves with increasing k. This is because all models used

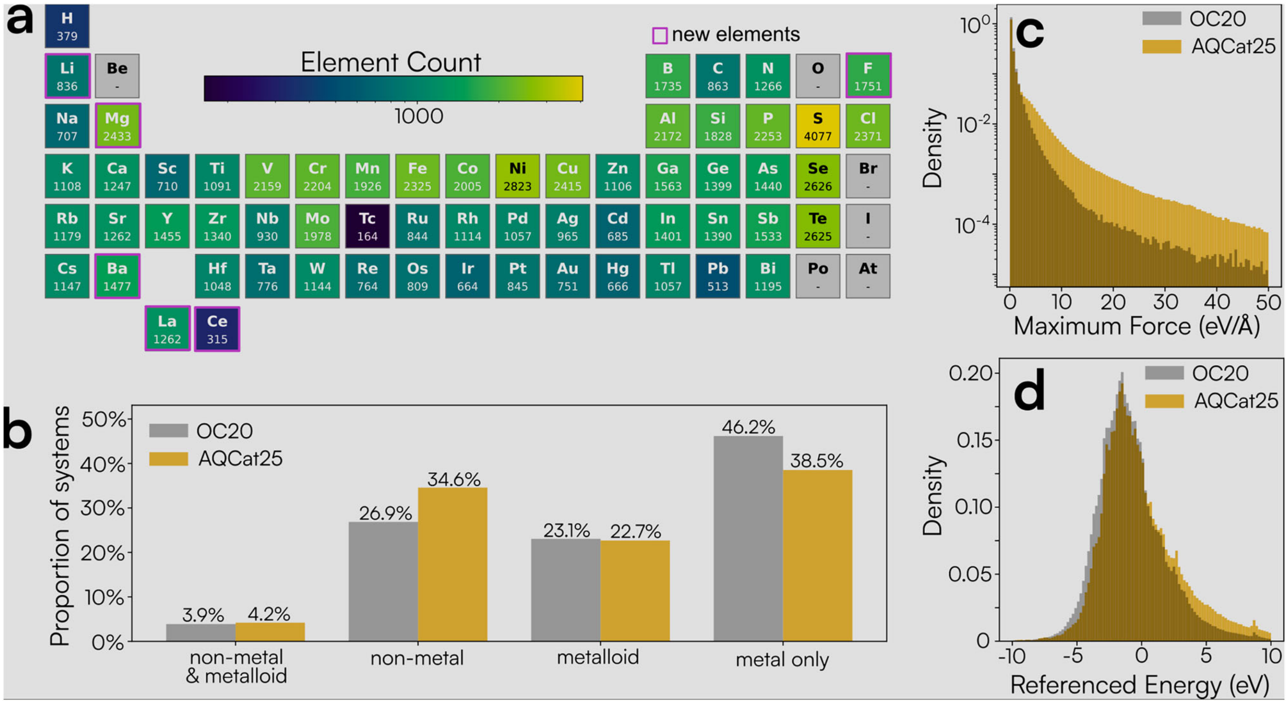

Fig. 2 | Comparison of AQCat25 and OC20 dataset statistics. **a** Element counts showing the frequency with which each element appears in the dataset. **b** The proportion of systems that fit into four material-type categories for the entire training splits of the OC20 dataset and AQCat25. **c, d** The distribution of adsorption energies and maximum forces across the training split of AQCat25 and the 2M training subsplit of OC20.

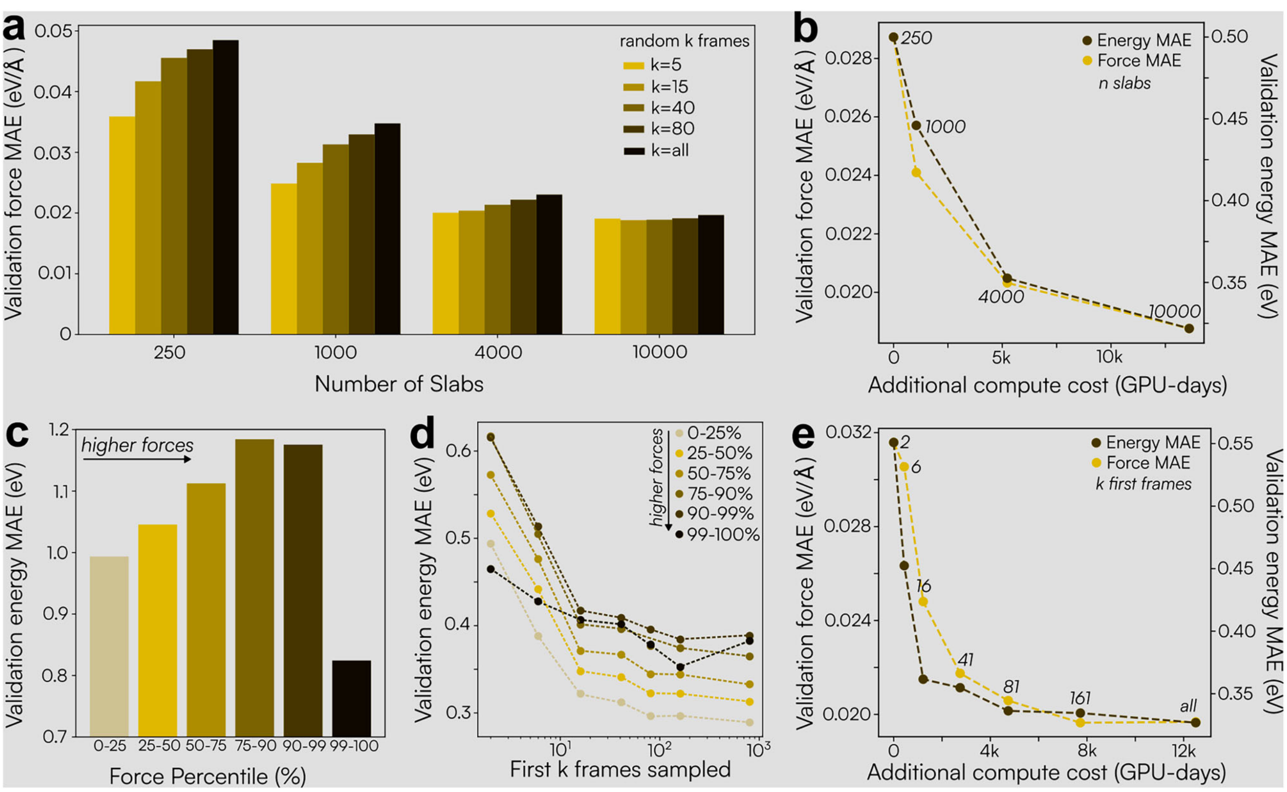

Fig. 3 | Optimizing cost reduction in model training and data generation. Exploration of trade-offs by directly fine-tuning 31M-parameter Equiformer v2 models. **a** The validation force MAE obtained when training models on random subsamples of the data for four dataset sizes. **b** The incremental compute cost for increasing the dataset size. **c** The pretrained 31M parameter Equiformer v2 model energy MAE on force percentile segregated data from the AQCat25 validation dataset. **d** Evolution in force percentile segregated energy MAE for fine-tuned models trained on variable first k frames from the dataset. **e** The trade-off between cost to generate training data and model performance when considering terminating relaxations after first k steps.

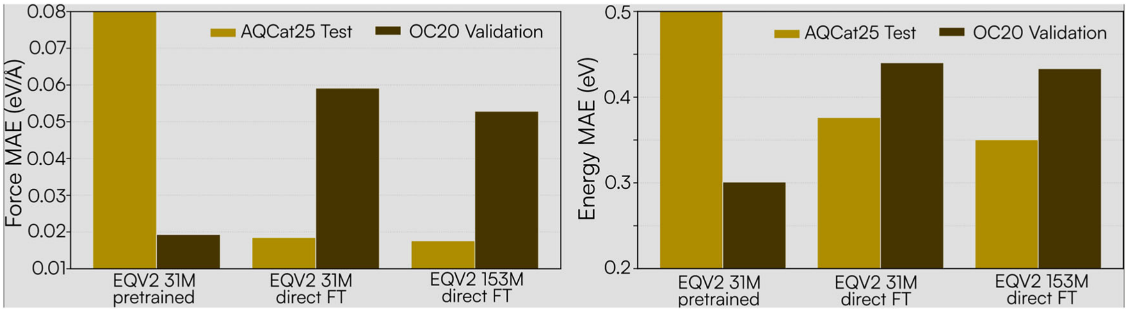

Fig. 4 | Performance of fine-tuned models on AQCat25 and OC20. Test force and energy MAE on AQCat25 test and a system-stratified subsample of the OC20 validation OOD, both split for three different models: a pretrained (on OC20 only) 31M parameter EV2 model, a 31M parameter EV2 directly fine-tuned (FT) on AQCat25, and a 153M parameter EV2 directly fine-tuned on AQCat25.

were trained using the very high force data (rattled and TS). Performance in this percentile is most influenced by high force data.

Using the first k relaxation frames presents an interesting trade-off between compute cost and model performance, which is captured in Fig. 3e. By only calculating between 40 and 80 frames instead of up to 800, we can achieve a Pareto optimum in model performance and compute cost for dataset generation. This would be our recommendation for future data generation campaigns. This exploration also revealed the advantage of training total energy models when designing a dataset to fine-tune models with cost in mind. If just 41 adsorbate-slab frames are computed, on average, 75% of the computation would be spent relaxing the slab completely. At 81 frames, this cost decreases to 63%, but it is still substantial. For total energy models, this cost is not necessary because relaxed slab energies are not needed to train. We also recommend designing datasets to train total energy models for future campaigns.

Model adaptation strategies

Although total energy models offer a clear path forward for more efficient data generation, this study focused on the adsorption energy target. Though arguably a more challenging learning task, as the model must implicitly account for the bare slab reference energy and any restructuring, adsorption energy may offer a significant convenience in established catalysis workflows. It also provides a well-defined target that isolates the adsorbate-surface interaction, which is ideal for developing a mixed fidelity/mixed physics adaptation protocol. Moreover, our overall objective is to create an MLIP that is broadly applicable, so we also wanted to understand model performance on the OC20 validation set. To assess the performance drift on the original OC20 task during these and subsequent experiments, we utilized a system-stratified subsample of the OC20 Val OOD Both split. This subset, which was sized to be comparable to individual AQCat25 validation splits (~300 k frames), is a computationally efficient metric for relative comparisons between models. The resulting MAE values, however, may not reflect absolute performance on the full OC20 distributions. An evaluation of performance for a pretrained 31M parameter EV2 model and two directly fine-tuned (FT) EV2 models with 31M and 153M parameters on the AQCat25 test and the subsampled OC20 validation split is shown in Fig. 4. We find that direct fine-tuning delivers reasonable AQCat25 errors, but deviation from the OC20 baseline on its validation split is significant. As anticipated, increasing model capacity from 31M to 153M parameters generally improves energy metrics on the AQCat25 test set. This is also true for increasing the energy loss weight (λ_E) (see Supplementary Table S1). The 153M model with $\lambda_E = 100$ yields the best energy MAE metrics in these direct fine-tuning experiments (Supplementary Table S1). However, the gains achieved by the larger 153M model may not justify its increased computational cost for practical applications, and this larger model still suffers from a significant performance drift for the original OC20 task.

To mitigate this drift, we explored opportunities to cotune and cotrain models using both subsamples of the AQCat25 dataset and the OC20 dataset. The models evaluated in this context, including those jointly trained with no additional OC20 data, incorporate the low-fidelity, spin-on data alongside the high-fidelity AQCat25 data. As seen in Fig. 5b, d, we observe that 0M models (models trained without OC20 data) exhibit a substantial deviation from the baseline OC20 performance (dashed black line). Unsurprisingly, the exclusion of the spin-off OC20 dataset leads to poor performance on OC20 validation relative to baseline metrics, even with the inclusion of the small low-fi spin-on set. Therefore, to produce a model that performs well across all domains (high/low fidelity, spin on/off) within a practical model size, we investigate cotuning and cotraining (from scratch) strategies that incorporate two amounts of the original OC20 data. Figure 5 summarizes the effect of including this OC20 data under two training regimes and three architecture variants.

Adding OC20 data consistently reduced deviation from the OC20 baseline on the examined validation split. For both cotuning and cotraining from scratch, the energy and force MAE trend toward the baseline as the amount of OC20 data increases for both energies and forces (Fig. 5b, d). We do not see this exact trend on the AQCat25 test split (Fig. 5a, c). In this case, for energies, we are showing the percent of frames that have an absolute energy error less than or equal to 0.2 eV. For this metric, a perfect model would have 100%. This was done because we observed that the energy values had strong outliers. One group of systems contributing to this phenomenon is those where the slab is organic (entirely composed of non-metals). This approach, as an alternative to MAE, is an unbiased way to ensure strong outliers do not skew the results. We have included some additional figures in the Supplementary Information to explore this metric further with different cutoffs (0.1, 0.3, 0.4, 0.6, and 0.8 eV instead of 0.2 eV) and using the MAE for the energies instead of the test and validation splits. It seems as though the results are sensitive to this metric, so we will only make broad conclusions. The percentages of errors within the threshold for AQCat25 energies are largely unchanged when increasing data for cotuning, whereas with cotraining, they increase (performance improves). For forces, there is a drastic increase in performance with more OC20 data for cotraining. For cotuning with FiLM, there is a modest improvement in forces when including OC20 data, but a substantial degradation when cotuning without FiLM. An economic approach when considering cost to train and performance on both AQCat25 test and OC20 validation is achieved when cotuning with 2M OC20 examples. Cotraining from scratch with FiLM improves performance for systems that have higher errors, though, so a tradeoff exists.

These patterns follow from standard behavior under distribution shift and multi-domain supervision. OC20 is broader and uses different DFT settings than AQCat25 (Supplementary Fig. S4). Fine-tuning only on AQCat25 moves the parameters toward that narrower distribution and

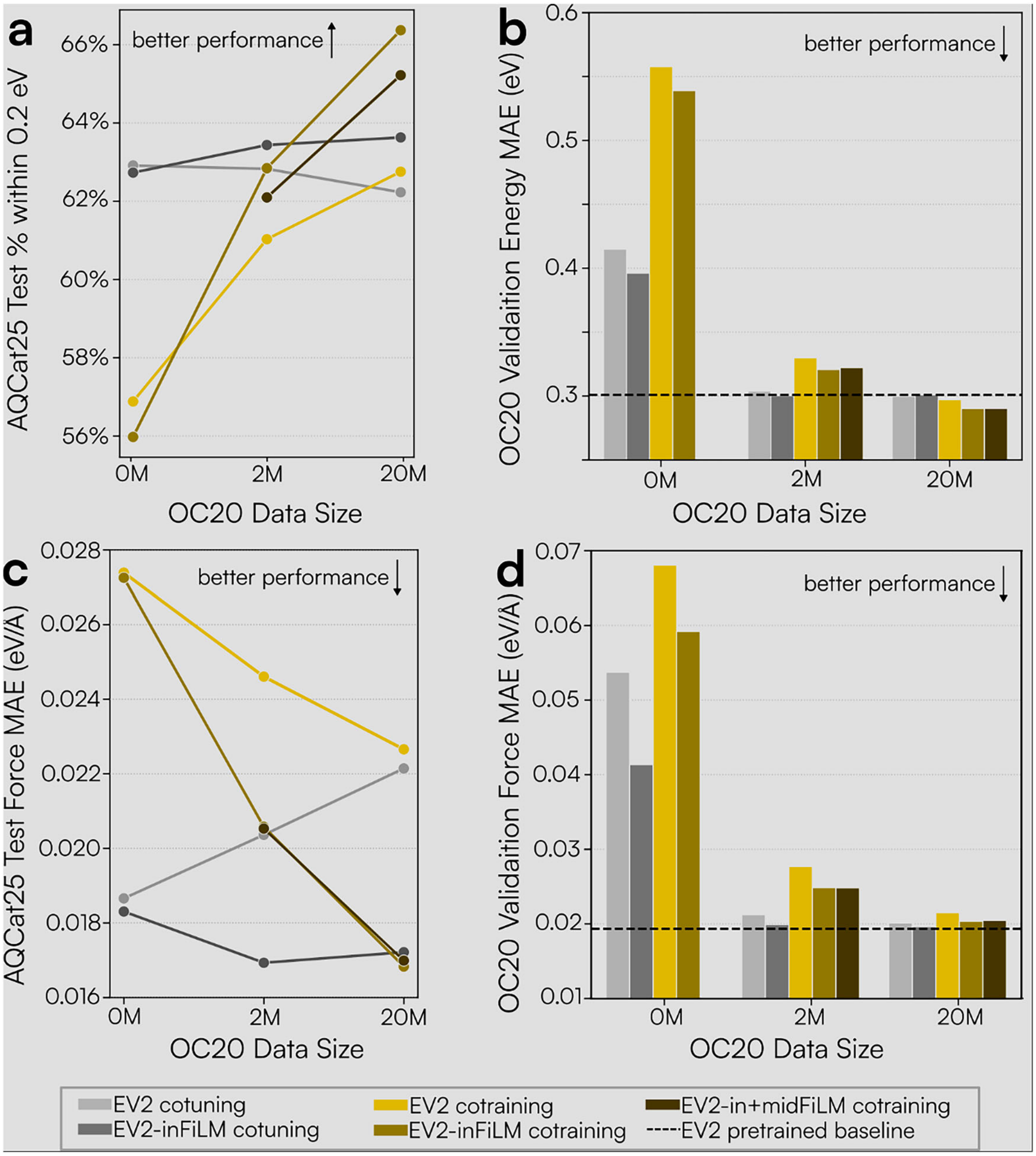

Fig. 5 | Cotuning and cotraining strategies. Model performance under cotuning and cotraining on AQCat25 and OC20. Part **a** shows the percentage of AQCat25 test set energies within 0.2 eV of the DFT value, while part **c** shows the force MAE on the AQCat25 test set, both as a function of the amount of OC20 data seen during training. Parts **b**, **d** explore the energy and force MAE trends on the OC20 validation set.

forgets OC20-specific features. Adding replay during cotuning without FiLM counteracts forgetting but also pulls the solution toward OC20 conventions, which explains the performance degradation on AQCat25 forces in Fig. 5c. Introducing FiLM provides a framework to distinguish these distributions, which rectifies the decrease in the force metric.

Starting from scratch changes the optimization path. The performance trends are strongly dependent on the OC20 data size used. Unsurprisingly, at 0M, jointly tuning clearly outperforms jointly training from scratch, but as OC20 data is added, their performance becomes sensitive to the metric. While tuning holds a slight advantage at a very strict 0.1 eV low-error threshold, the models cotrained from scratch show an advantage at higher cutoffs (e.g., 0.6–0.8 eV), indicating they are more effective at capturing outliers (see Supplementary Fig. S9a–f). FiLM makes the domain information explicit. Conditioning on spin and fidelity yields feature rescaling that reduces gradient interference between magnetic vs. non-magnetic and high- vs. low-fidelity cases.

Exploring robustness and generalization

To explore the robustness and generalizability of the models and form a more complete assessment of model usability, we constructed an additional validation set aimed at assessing the ability of the models to identify the global minimum energy for a given adsorbate-slab combination in line with the approach presented by Lan et al.[73]. We further explored differences in model performance when segmenting the data by interesting splits, namely the material type (metal-only, non-metal, metalloid, and metalloid+non-metal), whether spin was on or off, and whether the elements in a material were all included in OC20 or not.

The ultimate use of the MLIPs trained here will be for practical catalyst discovery, where an important figure of merit is the global minimum adsorption energy. We explored this using the dense DFT validation set with 50 relaxations each for 109 adsorbate-slab combinations. We performed ML relaxation inference starting from the same initial configurations as DFT. The relaxed states were filtered using the same algorithms presented by Lan

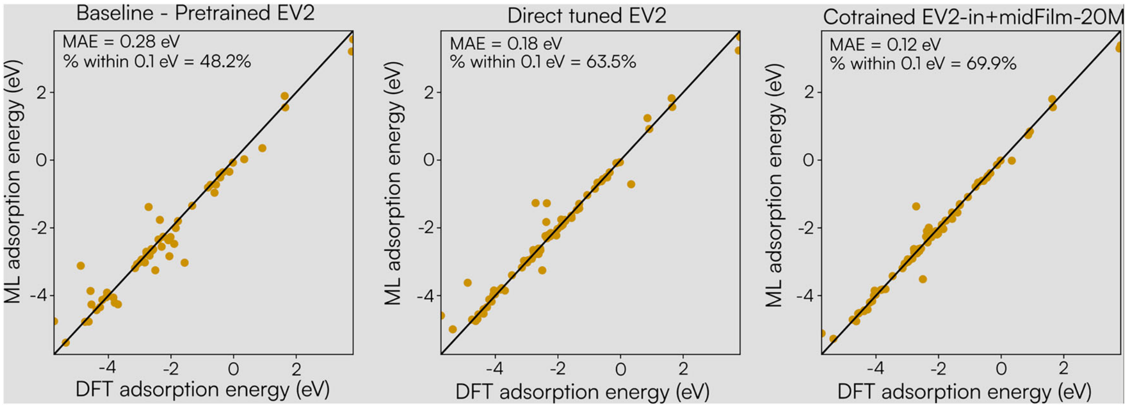

Fig. 6 | Adsorption energy prediction accuracy on a dense validation set. Parity plots between the DFT minimum adsorption energy and ML adsorption energy for a baseline pretrained 31M parameter EV2 model (left), the result of directly tuning that model on the AQCat25 dataset (center), and cotraining a 31M parameter EV2-in + midFilm model from scratch on both 20M examples from OC20 and the AQCat25 dataset (right).

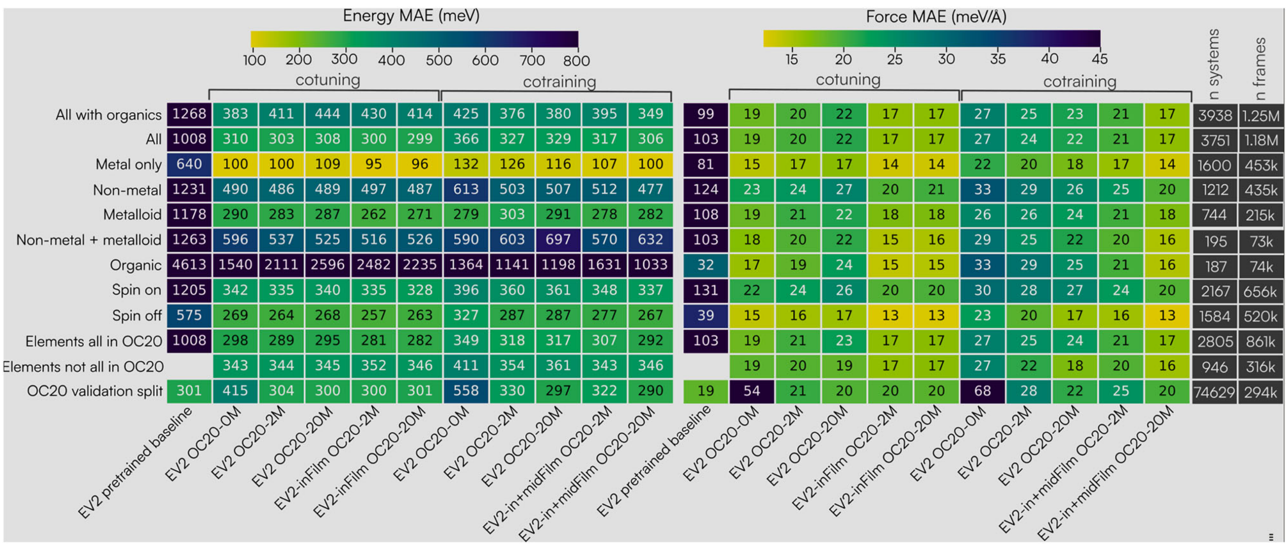

Fig. 7 | Analysis of error trends across chemical and physical domains. Comparison of Energy MAE (meV, left) and Force MAE (meV/Å, right) across different model training strategies and data subsets. Performance is evaluated for cotuning vs. cotraining, varying included OC20 data amounts (0M, 2M, and 20M), and different architectures (vanilla EV2, EV2-inFiLM, EV2-in + midFiLM). Subsets include spin treatment, element novelty relative to OC20, and material type.

et al.[73] to find desorption, dissociation, intercalation, and significant surface change. Figure 6 compares the performance of three 31M parameter models on this task, using only the ML-predicted energies without DFT single-points on the ML-relaxed structures. On the left is the pretrained 31M EV2 model (trained on OC20 All + MD), taken from the publicly available fairchem checkpoint. For this model, inference on systems containing new elements was omitted since performance would be poor. In the center is a directly fine-tuned EV2 model. On the right is the EV2-in + midFilM model, which was cotrained using 20M OC20 examples and the AQCat25 dataset. As a point of comparison, when the OC20dense[73] dataset was released, the EV2 model was not available. The best performing model was eSCN-MD-Large, and on this task, it had a 56.5% success rate with an energy MAE of 0.17 eV[73]. Here, the success rate is defined as the percentage of systems where the minimum adsorption energy found by ML is within 0.1 eV of the DFT value. ML success metrics alone were included in a later release as 60.8% for an EV2 model of unspecified size, 68.4% for the UMA-S model, 71.1% for the UMA-M model, and 74.4% for the UMA-L[55]. The success metric and MAEs have been annotated on the plots. We see the trend we would expect to see between models, with increasing performance from left to right. This further validates the usefulness of the models and also supports the fact that the loss function and training metrics are well designed and correlate with our downstream use case.

Evaluating material and magnetic subsplits

To further probe the robustness of the models presented and identify potential systematic biases related to specific chemical or physical properties, we next evaluate performance across distinct subsets of the test data. Specifically, we analyze error trends based on the material type, whether the elements contained were all included in OC20, and whether spin polarization was treated during the calculation. The results of this are shown in Fig. 7 with model energy MAE on the left and force MAE on the right. The data presented here are evaluated on the AQCat25 test split for all rows except the OC20 validation split, which is the same subsample of the OC20 out-of-domain (OOD), both split discussed above.

This analysis exposed a segment of the dataset that has very poor performance for energies: organic materials. Materials that only contain H, O, N, C, S, P, F, Cl, Br, I, and/or Se have very poor metrics as seen in the "Organics" row of Fig. 7. This poor performance, however, does not extend to forces. This is likely because of the referencing scheme used. These materials are more able to restructure, and it is therefore far more likely that

Table 1 | Stratification of model errors by magnetic and chemical domain across all hold-out sets

Category	Range (μ_B/at.)	E-MAE (meV)	E-Med (meV)	F-MAE (meV/Å)	F-Med (meV/Å)	% Metal	% Non-metal	% Metalloid	% NM/metalloid
Non-mag.	0−0.01	235	72	15.30	12.50	44.2	27.9	26.0	1.9
Weak	0.01−0.1	294	111	20.90	16.80	27.6	48.3	20.7	3.4
Moderate	0.1−0.5	389	173	23.00	17.90	22.9	58.2	15.3	3.6
Strong	0.5−1.5	312	130	21.30	16.30	24.9	50.8	21.2	3.1
Ultra-high	>1.5	221	79	21.70	16.90	42.4	20.4	32.8	4.4
Spin off	N/A	303	106	14.00	11.40	44.3	29.6	17.4	8.7

Comparison of mean and median errors across absolute magnetic moment ranges normalized by atom count (μ_B/at.) and the corresponding compositional distribution of each bin.

the relaxed slab state is very different than the adsorbate-slab along the relaxation trajectory. These materials are not necessarily of catalytic interest, so it could be beneficial to be more selective when including them in the dataset. Certainly, total energy models would be better suited to handle these systems because they remove dependence on the referenced state. Because of the significance of the errors on these systems, we removed them from all other splits presented in the figure. An alternative version of this figure has been included in the Supplementary Information where the organics are not segmented out. Without removing organics, we observe trends that are opposite to those expected and presented here.

Here, we see that across the board, model metrics for forces and energies are better for spin-off than spin-on. For most cases, when we compare the EV2 model to its corresponding EV2 + FiLM model, there is an improvement in spin on. Performance on systems where all elements appear in OC20 is better than on systems that contain at least one new element. Aligned with existing precedent, the models more accurately predict energies and forces on metals when compared to other material types. Metalloids are slightly better treated than non-metals. Interestingly, energies for non-metals + metalloids are worse than non-metals, but the forces are not.

The models jointly trained from scratch outperform corresponding jointly tuned models by ~1 eV on the challenging organic material split, but achieve slightly worse performance on the other material splits, excluding this category. The difference in optimization path and data exposure leads to these models marginally sacrificing performance on the broader set of materials to recover performance on this difficult class. Notably, as seen previously, the jointly trained from scratch model with 20M additional OC20 samples achieves the highest performance on the practical catalysis tasks described in Figs. 5a and 6.

To further disentangle the relationship between magnetic complexity and chemical composition, we stratified the model errors by both normalized magnetic density and material class (see Table 1). At first glance, the increase in Energy MAE from the Non-Magnetic regime (235 meV) to the Moderate regime (0.1–0.5 μ_B/at., 389 meV) suggests that spin polarization intrinsically drives model difficulty. However, the compositional breakdown reveals a more nuanced correlation: the Moderate regime is also the most heavily dominated by non-metal systems (58.2%).

Conversely, performance recovers markedly in the highest magnetic density regime (> 1.5 μ_B/at., 221 meV), strongly correlating with a shift toward metallic systems (42.4%). This suggests the elevated errors at intermediate magnetic moments are heavily compounded by the underlying chemical distribution. While intermediate magnetic states in non-metals often feature complex electronic topologies that may inherently challenge MLIPs, our data indicates model accuracy remains highly sensitive to the broader material class. However, we readily acknowledge that definitively disentangling the intrinsic difficulty of these magnetic states from their material chemistry will require further targeted ablations across isolated material classes.

A summary of model performance for the models discussed here, and some others, is shown in Table 2. Model names denote the architecture (EV2 31M default, inFiLM, or in+midFiLM) and training protocol, where "ft" signifies fine-tuning an OC20-pretrained model and its absence means cotraining from scratch. Dataset identifiers specify OC20 data added (+OC20-2M/20M). Direct tuning experiments excluded the low-fidelity spin-on subset.

Spin-branch ablation of adsorption thermodynamics and quenching

To systematically evaluate whether EV2-in + midFiLM-AQCat25+OC20-20M (shortened to AQCat25-EV2-FiLM for this discussion) encodes spin-mediated adsorption thermodynamics, a spin-branch ablation was performed on 14 systems from hold-out sets encompassing a range of normalized magnetic moments (μ_B/atom) and total magnetizations. These systems were randomly selected with the constraint being that all corresponding DFT calculations were treated with spin polarization enabled. The model was queried in two inference modes: (i) FiLM spin on, which activates the model's spin-aware features, and (ii) FiLM spin off, which does not, while holding the remaining configuration constant. The model branch gap ($E_{\mathrm{ads}}^{\mathrm{FiLM-Spin\ On}} - E_{\mathrm{ads}}^{\mathrm{FiLM-Spin\ Off}}$) serves as a proxy for the thermodynamic spin penalty. The macroscopic trends predicted by the model exhibit strong agreement with scaling relations established in the literature, which demonstrate that the chemisorption spin penalty scales proportionally with the substrate magnetic moment, strictly dependent on the radical character of the adsorbate[21,50].

For closed-shell adsorbates, represented by $^*H_2O$, the absolute value of the predicted model branch gap remains small across both non-magnetic and magnetic slabs, typically on the order of meV to tens of meV. This observation is consistent with literature demonstrating that while substrate d-band polarization induces a thermodynamic spin penalty for closed-shell molecules, the magnitude is fundamentally limited[50,74,75]. Theoretical benchmarks establish that the spin penalty for water is nearly zero on Ni surfaces and relatively minor, at approximately 0.15 eV, on Fe surfaces[50]. The model successfully replicates these modest, substrate-driven responses without over-penalizing the closed-shell species. Explicit density functional theory (DFT) validation on a subset (Supplementary Table S2) confirms this behavior, yielding true $^*H_2O$ branch gaps of <0.08 eV even on high-moment alloys such as Fe-Re.

In contrast, for open-shell radical *NO on strongly magnetic substrates, AQCat25-EV2-FiLM predicts substantially larger branch gaps. This aligns with findings by Nørskov and colleagues, who demonstrate that radical chemisorption on 3d magnetic transition-metal surfaces is systematically penalized relative to a theoretical non-spin-polarized state[50,74]. For radical-like species such as *NO or *NH, they report a thermodynamic spin penalty on Fe that is more than twice that observed on weaker magnetic substrates like Ni[21,50]. The observed branch gaps for AQCat25-EV2-FiLM adhere to this baseline and extend these expected scaling trends to more complex alloy systems (see Fig. S5 and Supplementary Table S2).

Although the primary scaling laws remain robust, the Fe_3N and FeGe-based systems exhibit distinct deviations from the linear trends. For Fe_3N, explicit DFT validation reveals an unpredicted stabilization of the adsorption state upon spin-polarization, contradicting the typical spin penalty expected. The discrepancy may arise from the model failing to accurately infer this specific composition (extracted from an OOD material distribution) or the reference DFT converging to a highly frustrated magnetic state.

Table 2 | Comparison of energy and force mean absolute errors (MAE) across different training strategies and validation sets

Category	Model	All E-MAE	All F-MAE	Spin On E-MAE	Spin On F-MAE	Spin Off E-MAE	Spin Off F-MAE	OC20 Val E-MAE	OC20 Val F-MAE
Pretrained	EV2-OC20	1268	98.63	1205	131.40	1378	37.06	301	19.32
Direct tuning	EV2-OC20-ft-AQCat25-highfi only	376	18.46	337	21.63	419	14.80	440	59.13
	EV2-OC20-ft-AQCat25-highfi only (153M)	350	17.59	339	20.41	362	14.34	433	52.84
Cotuning	EV2-OC20-ft-AQCat25	383	18.65	342	21.81	428	15.00	415	53.73
	EV2-inFiLM-OC20-ft-AQCat25	379	18.30	346	21.23	415	14.91	396	41.34
	EV2-OC20-ft-AQCat25 + OC20-2M	411	20.36	335	23.71	495	16.48	304	21.23
	EV2-inFiLM-OC20-ft-AQCat25 + OC20-2M	430	16.93	335	19.90	536	13.49	300	19.90
	EV2-OC20-ft-AQCat25 + OC20-20M	444	22.14	340	26.30	559	17.34	300	20.12
	EV2-inFiLM-OC20-ft-AQCat25 + OC20-20M	414	17.21	328	20.28	510	13.66	301	19.62
	EV2-inFiLM-OC20-ft-AQCat25 + OC20-20M (λ_E = 100)	412	21.03	325	24.32	508	17.22	289	21.79
Cotraining	EV2-AQCat25	425	27.38	396	29.86	457	24.53	558	68.06
	EV2-AQCat25 + OC20-2M	376	24.60	360	28.01	394	20.68	330	27.69
	EV2-inFiLM-AQCat25 + OC20-2M	392	20.57	345	23.79	442	16.86	321	24.86
	EV2-in+midFiLM-AQCat25 + OC20-2M	395	20.53	348	23.75	447	16.80	322	24.83
	EV2-AQCat25 + OC20-20M	380	22.65	361	26.85	402	17.81	297	21.51
	EV2-inFiLM-AQCat25 + OC20-20M	367	**16.83**	334	19.90	403	13.28	290	20.35
	EV2-in+midFiLM-AQCat25 + OC20-20M	**349**	16.98	337	20.05	363	13.44	290	20.46

Energy values are in meV and forces in meV/Å. The best performance for the fully joint-trained models is highlighted in bold.

Conversely, for the FeGe surface with *NO, explicit DFT validates the model's prediction of an anomalously high penalty (+0.85 eV ML vs. +1.01 eV DFT), which confirms that this system legitimately breaks the standard scaling relation through strong physical coupling rather than representing a model failure.

Conversely, the *NO radical undergoes strong covalent hybridization across all evaluated surfaces, evidenced by substantial charge transfer (CT > $0.60e^-$) and broadening of its molecular states across the Fermi level (Supplementary Fig. S6). Notably, the thermodynamic penalty associated with this covalent bond formation depends entirely on the substrate's magnetization. On $AgCu_4Sr$, *NO completely loses its radical character ($\Delta\mu_B = -0.74\mu_B$), which yields highly symmetric electronic states, in contrast with FeGe. According to the mechanism established by Nørskov and colleagues[50], this asymmetry drives a massive energetic penalty because the relationship between adsorption energy and the *d*-band center exhibits positive curvature. Consequently, the bonding gain from the upward-shifted minority-spin states is insufficient to offset the bonding loss from the downward-shifted majority-spin states, resulting in weaker overall adsorption.

Ultimately, these ablation results provide strong evidence that AQCat25-EV2-FiLM successfully distinguishes between the minor substrate-driven penalties characteristic of closed-shell adsorbates and the substantial energetic shifts associated with localized spin quenching when radicals hybridize with specific magnetic surfaces. Nonetheless, more systematic validations across a broader phase space of magnetic materials are necessary to fully map the boundary conditions and limitations of these learned spin-couplings.

Physical validity of learned conditioning

Finally, we probed whether the model's performance gains stem from genuine physical disambiguation or simple dataset-specific heuristics. This distinction is non-trivial; in the AQCat25 training set, magnetic elements (e.g., Fe, Co, and Ni) were exclusively calculated with spin polarization enabled, while in the OC20 portion of the training data, these same elements appeared exclusively in spin-unpolarized calculations. A naive model might simply associate these elements with a fixed energy offset or struggle to resolve the conflict between the two data sources.

To test this, we performed a qualitative check by analyzing the effect of toggling the spin context flag at inference time (Fig. 8) for a subset of OOD test samples. If the model had merely learned to apply a constant elemental bias to magnetic species, toggling the flag would likely yield uniform or negligible shifts. Instead, we observe a physically consistent response: forcing a hypothetical "spin off" flag for ferromagnetic systems correctly destabilizes them, resulting in a large positive energy shift consistent with the loss of exchange energy. Conversely, ground state non-magnetic systems show substantially lower energy changes when the flag is toggled (Nonmagnetic, Spin On/Off in Fig. 8), which suggests the model does not indiscriminately apply this same shift to all surfaces. Crucially, the model demonstrated the capacity to overcome incorrect training heuristics. In specific test cases where the dataset generation protocol defaulted to spin-off calculations despite a true magnetic ground state, as determined by the Materials Project (Ferromagnetic, spin-off—light gray in Fig. 8), enabling spin resulted in energetic stabilization.

The physically consistent energy shifts observed during this flag-toggling ablation suggest that FiLM-conditioned co-training allowed the model to internalize a switchable, structurally-aware representation of magnetism by contrasting the distinct physical regimes of OC20 and AQCat25. However, to provide transparent guidance on the boundaries of this physical learning, we must acknowledge the limitations imposed by the training data, particularly regarding complex anti-ferromagnetic (AFM) materials. These systems converged across a range from non-magnetic to strongly magnetic, with the vast majority settling into magnetic states (Table 3) due to our default spin initialization protocol.

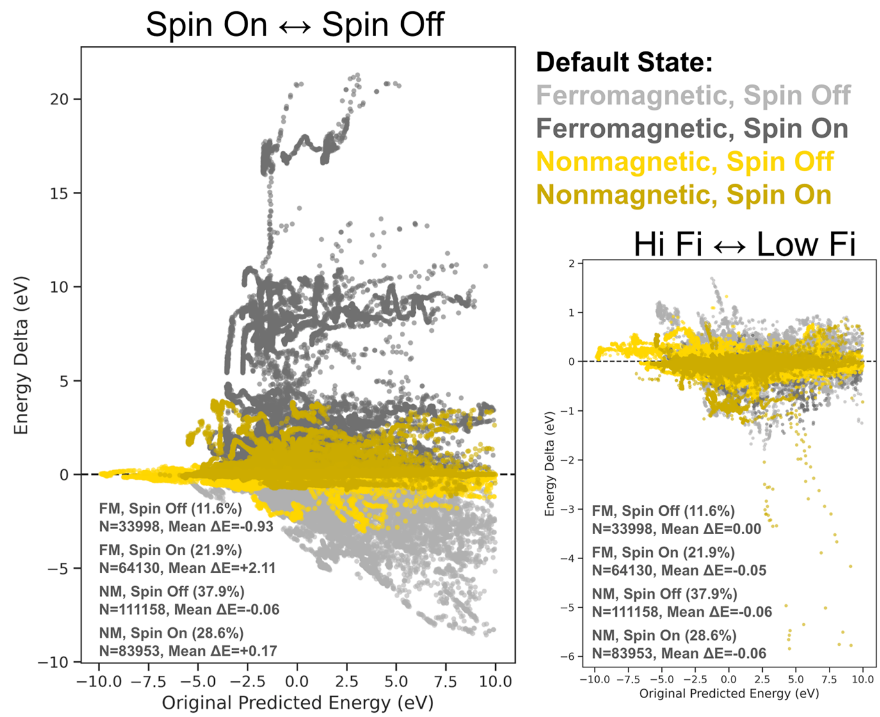

Fig. 8 | Physical validity of learned conditioning. Effect of toggling conditioning flags during inference on predicted energy delta (ΔE). Colors distinguish between ferromagnetic (FM, grey) and nonmagnetic (NM, yellow) systems. Toggling the spin flag induces large destabilizing energy shifts for FM systems while leaving NM systems largely unaffected. This suggests the model has learned to distinguish physical regimes rather than simply memorizing elemental biases, likely by leveraging the contrast between spin-polarized AQCat25 data and spin-unpolarized OC20 data for the same magnetic elements.

Table 3 | Proportions of unique adsorbate-slab combinations stratified by their Materials Project (MP) bulk ordering label and their final DFT-calculated magnetic density

MP bulk	Calculated adslab magnetic density (μ_B/at.)					
Ordering	Non-mag. (0–0.01)	Weak (0.01–0.1)	Moderate (0.1–0.5)	Strong (0.5–1.5)	Ultra-high (>1.5)	Total
AFM	8	15	42	128	13	206
FM	590	319	1671	2191	510	5281
FiM	43	44	508	809	62	1466
NM	16,172	3331	1958	62	5	21,528
Total	16,813	3709	4179	3190	590	28,481

Ranges for magnetic density are given in μ_B/at. AFM, FM, FiM, and NM denote anti-ferromagnetic, ferromagnetic, ferrimagnetic, and non-magnetic, respectively. While the bin ranges are heuristic, they capture distinct physical regimes: 0–0.01 accounts for true non-magnetic systems and standard DFT numerical noise, 0.5–1.5 encompasses mid-tier magnetic metals (e.g., bulk Ni is ~0.6 μ_B), and >1.5 μ_B captures ultra-high magnetic metals (e.g., bulk Co is ~1.7 μ_B, Fe is ~2.2 μ_B).

Because the model was trained on a phase space where stable AFM orderings are statistically negligible, users should interpret predictions on AFM materials with caution. In these regimes, the model's energy prediction likely represents the most statistically probable surface-relaxed state derived from the training heuristics, often a spin-quenched or generalized ferromagnetic state, which may not align with a verified AFM ground state. For the study of catalytic reactions on such complex surfaces, this foundational model is best utilized as an efficient pre-screening tool to rapidly generate candidate adsorption geometries and baseline energetic trends. Subsequent DFT calculations using explicit AFM initializations are recommended to accurately rank the final local magnetic ground states.

Ultimately, while our results indicate that physically-informed conditioning is an effective strategy for unifying conflicting physical approximations, we propose that future iterations could incorporate smaller, complementary seed datasets that systematically explore varied magnetic initializations. Fine-tuning foundational models on such data, while explicitly capturing site-specific magnetic moments as auxiliary model outputs, would provide the necessary physical constraints to learn spin competition. We posit this would enable the precise determination and ranking of predicted magnetic states without sacrificing the broad generalization capabilities learned during foundational pretraining.

Discussion

This work tackled a significant gap limiting the application of large-scale MLIPs in heterogeneous catalysis: the proper treatment of magnetism and enhanced electronic fidelity to accurately model and discover novel catalysts containing earth-abundant, spin-polarized elements such as Fe, Co, and Ni. We demonstrated that while direct fine-tuning of a pre-trained OC20 model on AQCat25 provides performance on the new data, it leads to a significant degradation of performance on the original OC20 domain. We found that by combining the targeted high-fidelity physics captured in AQCat25 with the extensive chemical and structural diversity present in a large portion of the OC20 data, jointly training successfully enhances accuracy on the AQCat25 test set while mitigating degradation on the evaluated OC20 validation metrics. We further confirmed the applicability of our models for the practical catalysis task of identifying the global minimum adsorption energy on a diverse set of surfaces. This training methodology, utilizing multi-fidelity data and explicit conditioning, offers a promising path toward practical and broadly applicable MLIPs for heterogeneous catalysis.

Methods

Density functional theory

DFT calculations were performed using the Vienna Ab Initio Simulation Package (VASP)[76–79]. A plane-wave cutoff energy of 500 eV was applied, and Gaussian smearing with a width of 0.1 eV was used. The revised Perdew-Burke-Ernzerhof (RPBE)[80,81] functional was chosen for its performance on heterogeneous catalyst systems, and the system's geometry was optimized using the conjugate gradient algorithm. The VASP parameters are summarized in Tables 4 and 5. The Bloch vectors (kpoints) were set using the lattice vectors using the same technique implemented in the fairchem repository for slabs and adsorbate-slab systems. The z-direction is set to 1, while x and y are set as $k = \max\left[\lfloor \frac{40}{c} \rceil, 1\right]$. For bulk, this calculation was also applied to the z-direction.

For systems containing Ce, Co, Cr, Cu, Fe, Mn, Mo, Ni, Os, Ru, V, or W, spin polarization was enabled to account for magnetic effects. Overall, the dataset consists of 49% spin polarized calculations. As detailed in Table 3, the magnetic ordering of the constituent bulk materials that constitute this dataset, according to the Materials Project (MP), is predominantly non-magnetic (NM) followed by ferromagnetic (FM). To bias the electronic solver toward the high-spin state during these calculations, initial atomic magnetic moments were assigned using consistent `Pymatgen` defaults (see Table 6 for specific values).

To evaluate the preservation of these bulk magnetic properties during surface cleavage and molecular adsorption, we cross-referenced the MP bulk labels with our final DFT-calculated adsorbate-slab magnetic densities (Table 3). We observe broad consistency, with the vast majority of bulk NM and FM systems retaining their respective non-magnetic or highly magnetic states. However, significant state migration also occurs. For instance, a subset of bulk NM materials (see Table 3) relax to weakly or moderately magnetic adsorbate-slabs, while over 11% of bulk FM systems exhibit spin-collapse to a non-magnetic state. These deviations from the bulk ordering are ascribed to the emergence of localized moments on specific facets due to broken dangling bonds, spin quenching induced by strong orbital overlap with polar adsorbates, and inherent differences between the MP bulk and AQCat25 adsorbate-slab computational settings. Consequently, the final dataset captures a diverse distribution of surface magnetic states that naturally diverge from idealized bulk priors.

Although these settings represent a significant increase in fidelity over previous large-scale datasets and are considered nominal for catalysis research, we acknowledge that even higher-fidelity calculations are possible. However, any increase in per-calculation fidelity must be weighed against the loss of dataset diversity for a fixed computational budget. For foundational MLIPs that must generalize across a vast chemical space, this trade-off is critical.

Table 4 | VASP parameters

Variable	Setting slabs systems	Setting bulks
IBRION	2	1
NSW	800	250
ISIF	0	7
ISPIN	1 or 2	1 or 2
ISYM	0	0
ALGO	Normal	Normal
ISMEAR	0	0
SIGMA	0.1	0.1
EDIFFG	-0.03	1E-5
ENCUT	500	500
PREC	Accurate	Accurate
POTIM	0.5	0.5
NELM	250	250
EDIFF	1E-4	1E-4
SYMPREC	1E-10	1E-5
LREAL	Auto	False

Table 5 | MD specific VASP parameters

Variable	Setting
TEBEG	900
TEEND	900
MDALGO	1
ANDERSEN PROB	0.0
NSW	80
POTIM	2
IBRION	0
NELMIN	4

Adsorbate referencing

The adsorbate gas phase references were constructed using the energies of CO, H_2, H_2O, and N_2. Calculations were performed with the molecules separately in vacuum cubes of 10, 20, and 30 Å. There was not a significant energy difference between 20 and 30 Å, so 30 Å was taken to be converged. The resulting per-atom/element energies are summarized in Table 7.

Bulk selection

The AQCat25 bulk materials database was constructed by first updating and then expanding the OC20 dataset. Initially, the MP[82,83] database was queried for all structures containing only elements present in the OC20 dataset, subject to the constraints of a maximum of three unique elements per material and an energy above the convex hull (E_{hull}) of 0.1 eV/atom or less. Structures from the original OC20 dataset not found in this query were retained only if they were structurally unique. To expand the chemical space, a second query was performed using the same stability and size constraints

Table 6 | Initial magnetic moments for spin-polarized DFT calculations

Element	Initial *MAGMOM* (μ_B)	Element	Initial *MAGMOM* (μ_B)
3*d* Transition metals		4*d*/5*d* Transition metals	
V	5.00	Mo	5.00
Cr	5.00	Ru	2.20
Mn	5.00	W	5.00
Fe	5.00	Os	2.20
Co	5.00		
Ni	5.00	Lanthanides	
Cu	1.73	Ce	5.00

Values represent the starting magnetic moment (μ_B) assigned to each atom of the specified element at the beginning of the electronic relaxation, acquired from `Pymatgen` defaults. All other elements were initialized with a magnetic moment of 0.0 μ_B.

Table 7 | Adsorbate per atom energy corrections

Atom	Energy [eV]
H	−3.4944
O	−7.1590
C	−7.2654
N	−8.1351

Table 8 | The number of systems and single points across data splits

Primary split	Secondary split	*N* systems	*N* single points
	Relaxations	24624	6959 k
In domain	Rattled	8189	947 k
	Transition states	2854	676 k
	Molecular dynamics	2098	249 k
	OC20 fidelity, spin on relaxations	4831	863 k
	OOD adsorbate relaxations	1913	577 k
Validation	OOD material relaxations	991	318 k
	OOD both relaxations	994	295 k
	OOD adsorbate relaxations	992	347 k
Test	OOD material relaxations	994	316 k
	OOD both relaxations	988	356 k
	ID	19,273	1282 k
	ID OC20 fidelity, spin on	4868	273 k
Slabs	OOD validation	497	29 k
	OOD test	498	36 k
Totals		47 k	13.5 M

The total system count reflects the number of unique adsorbate-slab combinations.

but including six additional elements: Li, Ba, La, Ce, Mg, and F. The resulting set of new materials was then subsampled to ensure balanced representation. Up to 500 structures were randomly selected for each group containing a single new element, and up to 20 structures for each group containing a combination of new elements. The dataset was assembled by combining the updated OC20 dataset materials, the preserved unique structures, and the sampled new materials. Finally, the dataset was filtered to only contain bulks with up to 30 atoms per unit cell. Data splits were then assigned by attempting to preserve the original designations for all OC20 dataset materials and distributing new materials based on chemical composition to maintain consistency with the established OC20 dataset splitting methodology.

Adsorbate-slab selection

The number of single points and systems that make up the splits and data types included in the AQCat25 dataset is shown in Table 8. Here, a system is defined as a unique adsorbate-slab pair for that subsplit.

Dataset splits

The dataset is structured into three primary splits: in-domain (ID), OOD validation, and OOD test. Each OOD split is further categorized by the type of novelty introduced, either in the adsorbate or in the material slab. This strategy is designed to evaluate the model's ability to generalize to novel systems it has not seen during training, in the same manner as the OC20 dataset[36]. The ID split contains configurations where both the adsorbate and the material slab are present in the training set. The test and validation ID splits serve as a baseline for the model's performance on familiar data and are sampled from the same distribution of the training set. Both OOD splits are designed to test the ability of machine learning models to generalize. The OOD validation set is used for hyperparameter tuning, while the OOD test set provides a final, unbiased evaluation of the model's performance on unseen data.

The following categories are included in both OOD splits: (1) OOD adsorbate, (2) OOD material, and (3) OOD both. For the OOD adsorbate, the material slab is ID, but the adsorbate is new and does not appear in the training data. The test OOD adsorbates also do not appear in the validation split, and the validation OOD adsorbates also do not appear in the test split. For OOD material, the adsorbate is ID, but the bulk lattice structure (not necessarily its composition) used to construct the slab is new and does not appear in the training set. For OOD, both the adsorbate and the material slab are new and do not appear in the training set. The same segregation for validation and test also applies.

Sampling diverse states

To ensure models trained with this dataset have a robust understanding of different structural and energetic states, we employed several calculation types for data generation. The dataset samples both high-energy, off-equilibrium states and low-energy, near-equilibrium states. To sample low-energy states, we performed adsorbate-slab structure relaxations. Relaxation calculations involve iteratively optimizing atomic positions to find a local energy minimum. A DFT call is made to determine forces, and atoms are moved according to the conjugate gradient method. This process is repeated until the maximum force on any atom is less than 0.03 eV/Å or a maximum of 800 steps are reached. These trajectories sample a range of configurations from high to low forces. All OOD validation and test set calculations are relaxations.

To sample high-energy configurations of adsorbate-slab systems, we took three approaches: (1) running MD calculations, (2) placing TS-like systems, and (3) rattling atoms. MD calculations were started from a relaxed adsorbate-slab structure, and run for 80 steps of MD at 900 K. To provide the model with examples of highly distorted configurations relevant to chemical reactions, we extracted transition state structures from the OC20NEB dataset[84]. These adsorbates were placed on new surfaces, followed by a short 5-step relaxation. This process generates data with high forces and energies, supporting the training of models that can handle reactive states. To further augment high-force data, we generated rattled configurations by randomly perturbing atomic positions. Two methods were used: (1) rattling all atoms and (2) rattling only adsorbate atoms, with displacements sampled from a normal distribution ($\sigma = 0.05$, 0.1, 0.15, or 0.2 Å). Some rattled systems underwent a single DFT calculation, while others had a short 5-step relaxation. Systems whose max absolute force or absolute adsorption energy exceeds 50 eV/Å and 10 eV were excluded from training and evaluation. We note that alternative efficient data generation strategies could further enhance diversity by utilizing machine learning molecular dynamics (MLMD) simulations to generate unique configurations for subsequent high-fidelity DFT labeling. This approach was not pursued here due to the initial lack of a reliable foundational model for spin-polarized catalyst surfaces, but it represents a promising frontier for future dataset expansions.

Additional data

To explore the opportunity to train models with less costly DFT data, we considered data that include spin polarization but with settings that otherwise match the OC20 dataset[36]. Notable differences between this data and the rest of the AQCat25 dataset are that we used a plane wave cutoff of 350 eV and Methfessel-Paxton smearing with a width of 0.2 eV. This data aids in understanding how the model handles the distinct physical regimes defined by fidelity and spin polarization. This dataset complements the existing high-fidelity spin-on/off (AQCat25) and spin-off OC20 data by filling a missing quadrant. Adsorption energies were computed using high-fidelity adsorbate references for all spin-on systems. We found this to have little impact on the final target energies from preliminary tests.

We also wanted to form an understanding of model performance on the task of finding the minimum adsorption energy for an adsorbate-slab combination. To do this, we constructed a small dense dataset, similar to the OC20dense dataset presented by Lan et al.[73]. For this dataset, we selected 109 adsorbate-slab pairs. Adsorbates were selected to be dissociation reactants from the OC20NEB dataset[84]. Slabs were selected randomly, but we selected the materials they were cut from more strategically. We included five unary materials, five binary non-metal materials, 46 binary intermetallics, 30 ternary intermetallics, and 23 ternary non-metals. Within these categories, the bulks were also randomly selected from the bulk database. For each adsorbate-slab pair, we performed 50 placements using the random site with heuristic placement mode in `fairchem`[85]. These placements were relaxed with the same DFT settings as the broader AQCat25 dataset. The relaxed states were filtered using the same algorithms presented by Lan et al.[73] to find desorption, dissociation, intercalation, and significant surface change.

System enumeration

All systems were prepared using the publicly available `fairchem` package[85]. Slab enumeration was performed using the underlying `pymatgen`[86,87] algorithm. Adsorbate placement was performed heuristically at random sites. Rattled systems were perturbed after adsorbate placement using the rattle functionality in `ASE`[88,89]. For TS systems, they were placed as normal adsorbed intermediates would be by preparing a new adsorbate database with TS entries.

Machine learning experiments

A challenge in this work is training a single MLIP that can accurately predict energies and forces across a dataset containing multiple DFT settings. The combined training data spans four distinct physical regimes: high-fidelity spin-on, high-fidelity spin-off, low-fidelity spin-on, and the original low-fidelity spin-off. We therefore explored methods to introduce this DFT context, namely spin treatment and calculation fidelity, directly to the model. Inspired by the effectiveness of Feature-wise Linear Modulation (FiLM)[90] and similar successful techniques[55] for multi-task learning, the baseline models presented in this paper focus on adapting the EquiformerV2 (EV2) architecture[53] using this approach. We did not focus on fine-tuning UMA[55] models because of licensing issues, but we expect those to have even better performance than the models presented here. The Feature-wise Linear Modulation (FiLM) technique[90] provides an expressive conditioning mechanism. Rather than adding context via feature concatenation, FiLM applies a learned, feature-wise affine transformation ($\gamma F + \beta$) that can scale, shift, or suppress activations. This strategy of deep, additive modulation is similar in principle to the additive embedding mechanism successfully employed by the UMA family of models[55]. To evaluate performance on AQCat25 while retaining knowledge from OC20, we used the EV2[53] model architecture with three variants and three training protocols. The variants were: (i) *EV2* (unmodified), (ii) *EV2-inFiLM*, which applies additive FiLM[90] shifts to the scalar ($l = 0$) channels at the input, and (iii) *EV2-in + midFiLM*, which applies the same modulation at the input and after each equivariant block. The protocols were: direct fine-tuning of OC20-pretrained checkpoints, cotuning those checkpoints with OC20 replay, and cotraining from scratch on mixed data from both AQCat25 and OC20.

It is important to note how baseline performance was assessed in this context: evaluations of pretrained models on AQCat25 used the provided structures directly, without re-optimizing lattice constants using OC20 DFT settings. These metrics represent the performance inherited for subsequent tuning rather than the inherent capability of the OC20 model on these specific materials had geometries been fully relaxed with consistent settings. Similarly, all evaluations performed on the OC20 validation subset utilized structures with OC20-optimized lattice constants.

Machine learning architecture

EV2 is an E(3)-equivariant transformer over atomic graphs: atoms form nodes, edges use pairwise distances and spherical harmonics, attention layers are equivariant, and feed-forward layers use S^2 activations[53,91–93]. Energies are predicted by a scalar head, and forces are predicted directly via a vector head. Architectural hyperparameters are listed in Table 9.

FiLM conditioning supplies the network with compact context about the DFT settings of each structure. Two binary indicators encode spin treatment and fidelity (`spin_on` and `low_fi`). Each indicator is embedded; the embeddings are concatenated and passed through a small multilayer perceptron (MLP) to produce a modulation vector β. We apply β additively to the scalar channels broadcast across nodes (and per block for EV2-in + midFiLM). Preliminary tests with multiplicative factors γ did not improve validation metrics measurably, so we retained the additive shift only for simplicity. Figure 9 diagrams the module and its insertion points.

Table 9 | Architectural hyperparameters for the EquiformerV2 models, including FiLM specifics

Hyperparameter	Value
Core EquiformerV2 architecture	
Number of transformer blocks	8 (31M), 20 (153M)
Embedding dimension d_{embed}	128
$f_{ij}^{(L)}$ dimension d_{attn_hidden}	64
Hidden dimension in feed forward networks d_{ffn}	128
Number of attention heads	8
Maximum spherical harmonic degree (L_{max})	4 (31M), 6 (153M)
Maximum spherical harmonic order (M_{max})	2 (31M), 3 (153M)
Dropout rate	0.1
Stochastic depth	0.1
Cutoff radius (Å)	12.0
Maximum number of neighbors	20
FiLM architecture addendum (EV2-FiLM)	
Auxiliary feature embedding dimension	16
MLP hidden dimension for modulation	128
MLP dropout	0.1
FiLM modulation strategy	*Cotuning*: -Input layer only *Training from scratch*: -Input layer only -Input layer and all transformer blocks

We refer to the EquiformerV2 paper[53] for a complete description of all architectural components, including normalization and activation functions.

We note this conditioning framework is inherently extensible to higher-dimensional physical contexts beyond binary indicators. Provided with more diverse training data, future iterations could explicitly encode a range of computational parameters and site-specific magnetic initializations to enable precise interpolation across electronic and magnetic regimes. Further, benchmarking on an H100 GPU confirms that the FiLM module introduces negligible memory overhead (<1 MB) with the *inFiLM* model showing effectively no speed penalty (within ±2%) relative to the vanilla model. The *in + midFiLM* variant exhibits a modest 8–17% reduction in katoms/s due to the additional read/write operations required for intermediate layer conditioning.

Training and adaptation protocols

To test the model's adaptation to the AQCat25 domain under distribution shift via direct fine-tuning, we started from public EV2 OC20 All+MD checkpoints (31M and 153M parameters) and fine-tuned on AQCat25 directly. For cotuning with OC20 replay, we fine-tuned the OC20 checkpoints on a composite stream consisting of AQCat25 high-fidelity and, when specified, a small AQCat25 low-fidelity spin-on stream, mixed with OC20 spin-off data at 0M/2M/20M scales. Finally, for cotraining from scratch, we trained EV2, EV2-inFiLM, and EV2-in+midFiLM from random initialization on the same composite streams used for cotuning.

Hyperparameters, controls, and compute

Optimization and architectural settings are summarized in Tables 9 and 10. For each training run, 8 × H100 NVIDIA GPUs were used. Unless noted, the force term dominated the objective; the default loss ratio was λ_E:λ_F = 4:100. We did not apply explicit loss-weighting penalties or domain-specific rebalancing based on the spin flag. Spin-on and spin-off data were treated equally in the loss calculation, as we found the FiLM conditioning architecture naturally arbitrates gradient contributions between domains without manual scaling. To reduce the computational cost for the extensive model adaptation experiments, the AQCat25 dataset component was subsampled.

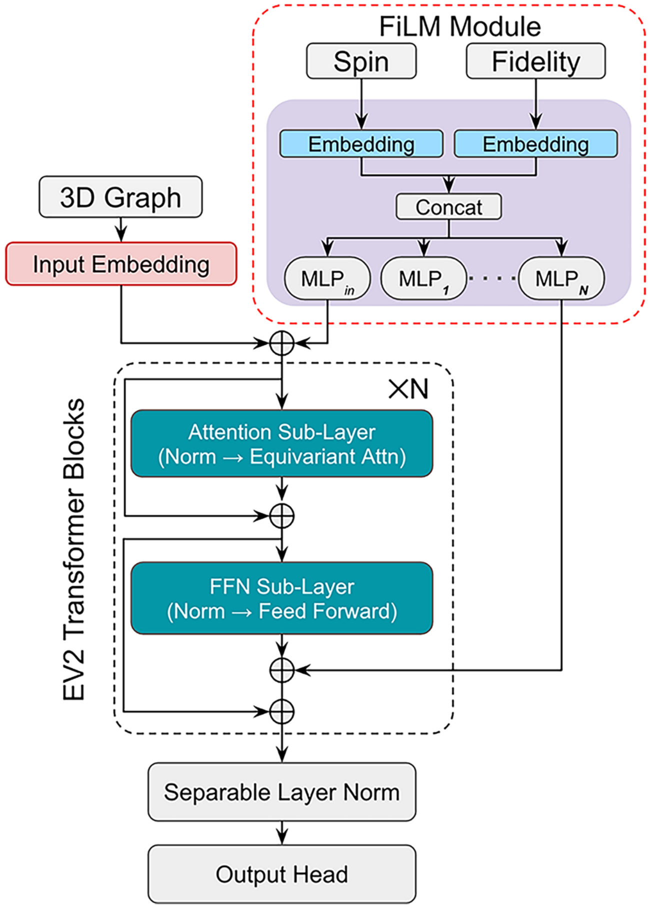

Fig. 9 | FiLM module architecture. Binary context (spin, fidelity) is passed through embeddings and an MLP to produce β. This β additively modulates scalar channels at the input (inFiLM) and optionally mid-block (in + midFiLM).

Table 10 | Training and optimization hyperparameters for each experimental strategy

Parameter	Direct finetuning	Cotuning	Training from scratch
Pre-trained checkpoint	OC20 All + MD	OC20 All + MD	None
Optimizer			
Optimizer	`AdamW`	`AdamW`	`AdamW`
Weight decay	1×10^{-3}	1×10^{-3}	1×10^{-3}
Learning rate (31M)	7×10^{-5}	7×10^{-5}	4×10^{-4}
Learning rate (153M)	8×10^{-5}	8×10^{-5}	4×10^{-4}
LR scheduling	Cosine annealing with linear warmup		
Warmup epochs	0.01	0.01	0.1
Model EMA decay	0.999	0.999	0.999
Batch size & epochs			
Batch size per GPU (31M)	20	20	20
Batch size per GPU (153M)	6	6	6
Gradient accumulation (153M)	3 steps	3 steps	3 steps
Effective batch size (31M)	160	160	160
Effective batch size (153M)	144	144	144
Max epochs	30	30	30
Loss and regularization			
Energy coefficient (λ_E)	4, 100	4	4
Force coefficient (λ_F)	100	100	100
Gradient clipping norm threshold	5	5	100

Models were trained in single precision for a consistent comparison across the numerous experimental conditions and ablations. For tuning experiments, we tested stronger regularization by increasing weight decay and by lowering the learning rate; both choices produced early plateaus and higher validation errors on AQCat25 relative to fully thawed baselines. Except for models trained from scratch solely on AQCat25, energy and force targets were normalized using mean/standard deviation values from the OC20 distribution, as this yielded slightly improved performance in preliminary tests. We also tried incremental thawing schedules that kept the backbone frozen while adapting only input embeddings (to accommodate new elements), followed by gradual unfreezing. These schedules underperformed fully thawed tuning on AQCat25.

Additionally, we explored alternative conditioning mechanisms and found that a simpler baseline involving direct concatenation of context embeddings performed competitively with FiLM when cotuning with limited (2M) OC20 data replay. We further experimented with more complex architectural modifications aimed at adapting the pretrained weights, including adding separate prediction heads routed by the conditioning flags and incorporating lightweight adapter modules within the transformer blocks. However, these approaches did not yield significant performance enhancements over the FiLM-based conditioning and fully thawed training strategies presented here. Finally, we do not claim hyperparameter or schedule optimality. Alternative warmup/decay, replay curricula, batch-composition policies, weight decay, EMA, or gradient clipping may yield further gains. Our goal here is a consistent and reproducible setup that enables clear comparison across regimes and architectures for the baseline models being presented.

We note that the binary FiLM conditioning approach is not meant to disambiguate between multiple degenerate or near-degenerate magnetic orderings (e.g., ferromagnetic vs. anti-ferromagnetic vs. low-spin states) for a single atomic configuration. As the current study relies on a high-throughput data generation protocol with consistent initializations (see Table 6), the spin-polarized subset of AQCat25 is likely biased toward ferromagnetic ground states (see Table 3). Consequently, the model's predictions are expected to reflect the underlying magnetic heuristics of the training data. Unlike recent specialized architectures that utilize explicit magnetic moment inputs to rank magnetic orderings with high precision[47], our approach focuses on providing a scalable framework for cross-domain learning between large-scale unpolarized (OC20) and polarized (AQCat25) datasets. By treating the spin flag as a global physical context rather than a local degree of freedom, our goal is to enable the model to distinguish between distinct physical approximations (spin-on vs. spin-off) across a vast chemical space.

While this binary conditioning is an initial effort toward a large-scale magnetic dataset for heterogeneous catalysis, it establishes a foundational architecture that can be further refined with additional data. We propose that future iterations could resolve current limitations by incorporating smaller, complementary seed datasets that systematically explore varied magnetic initializations. Fine-tuning foundational models on such data, while explicitly capturing site-specific magnetic moments as auxiliary model outputs, would provide the necessary physical constraints to learn spin competition. This would enable the precise determination and ranking of predicted magnetic states without sacrificing the broad generalization capabilities learned during foundational pretraining.

Spin and fidelity ablation calculations

To investigate the impact of spin and fidelity on the resultant energies (as shown in Supplementary Fig. S4), we performed DFT single points on the DFT relaxed (with AQCat25 settings elections) adsorbate-slab configuration and the DFT relaxed slab. The energies are presented as the difference between these two energies to exploit a cancellation of error from any differences in the true lattice constant. The single points were performed specifically by ablating the settings highlighted. For fidelity, 500 spin-on systems and 500 spin-off systems were selected, and single points were performed with a plane-wave cutoff of 350 eV, and Methfessel-Paxton smearing with a width of 0.2 eV. For spin, 1000 systems with spin on were selected, and single points were performed with spin off. For both spin and fidelity, 1000 systems with spin on were selected, and single points were performed with the alternative fidelity and spin off. Additionally, CT was computed via Bader analysis as the net difference between the integrated valence electron count on the adsorbed molecule and its neutral, gas-phase valence.

Data availability

The AQCat25 dataset is publicly available on Hugging Face at https://huggingface.co/datasets/SandboxAQ/aqcat25-dataset. Model checkpoints, including the 31M and 153M EV2 variants, are available at https://huggingface.co/SandboxAQ/aqcat25-ev2. All artifacts are provided under the Creative Commons Attribution Non-Commercial Share Alike 4.0 International (CC BY-NC-SA 4.0) license. Detailed instructions for data access and model inference are provided in the repository documentation.

Code availability

The training code corresponding to the models, including the specific FiLM conditioning implementation, is publicly available within the model repository at https://huggingface.co/SandboxAQ/aqcat25-ev2. Detailed instructions for model inference are provided in the repository tutorials.

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Author contributions
O.A. performed model implementation, experimental design, training, ablations, idea conceptualization, data processing, data analysis, data visualization, writing, and editing. B.W. performed dataset generation, idea conceptualization, data analysis, data visualization, writing, and editing. A.R.S. provided project leadership, idea conceptualization, writing, and editing. S.K., R.P., P.A. and A.N. managed the computing infrastructure. T.S., J.C., T.L., J.K., R.W., S.A., A.R., A.F., C.W., A.W., T.M. and K.R. contributed to code validation and manuscript review. All authors have read and approved the manuscript.

Competing interests
The authors declare no competing interests.

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